

Francesca Izzì

E-mail: francesca.izzi91@gmail.com

LinkedIn: francesca-izzi



Profile Summary

R&D Scientist with 6+ years of hands-on laboratory experience in protein biochemistry, mammalian cell culture, recombinant protein expression, affinity purification, protein-interaction assays, and proteomics-related workflows. Experienced in assay optimization, quantitative data analysis, mass spectrometry sample preparation, and SOP-based experimental work, with a strong focus on reproducibility and data quality. Interested in expanding this background into protein bioconjugation and characterization for ADC development.

Work experience

12.2023 – 05.2025

Postdoctoral Scientist

Deutsches Zentrum für Neurodegenerative Erkrankungen e.V. (DZNE)
Tübingen, Germany

- Developed, optimized, and validated quantitative protein-interaction assays (proximity labeling, genetic knock-down).
- Ensured data reliability and reproducibility through SOP-based workflows and structured documentation.
- Prepared analytical summaries for internal reports and multi-group discussions.

11.2021 – 06.2023

Postdoctoral Scientist

Deutsches Zentrum für Neurodegenerative Erkrankungen e.V. (DZNE)
Tübingen, Germany

- Executed and optimized biochemical and cell-based assays for protein-interaction and inhibitor studies.
- Performed quantitative data analysis (GraphPad Prism, R) and contributed to cross-functional projects.

10.2017 – 12.2020

Research Assistant

Natural and Medical Science Institute at the University of Tübingen (NMI)
Reutlingen, Germany

- Developed neuron-glia co-culture assays on microfluidic platforms with electrophysiological readouts.
- Validated methods for reproducible quantitative assays and maintained SOP documentation.

Education

10.2017 – 09.2021

PhD Cellular and Molecular Neuroscience, Graduate Training Centre of Neuroscience, University of Tübingen, Germany

10.2014 – 12.2016

Master of Science Pharmaceutical Biotechnologies
Università degli Studi di Padova, Italy

10.2010 – 09.2014

Bachelor of Science Biotechnologies
Università degli Studi di Padova, Italy

Technical skills

Cell Biology	Primary cell isolation and culture (murine DRG neurons), standard cell lines (HEK293, A549), cell isolation (MACS), cell transfection, microfluidics
Molecular Biology and Biochemistry	Immunocytochemistry, live-cell imaging, gel electrophoresis (SDS-PAGE), Western Blot, Jess, protein expression, protein extraction, protein purification, BCA assay, sample preparation for MS
Analytical and Biophysical Methods	Electrophysiology (MEA, MF-MEA), chromatographic methodologies (HPLC; familiar with FPLC/ÄKTA workflows for downstream purification), mass spectrometry (ESI), Ussing chamber
Imaging	Confocal fluorescence microscopy, light microscopy

Data analysis and software

Advanced	MS Office, GraphPad Prism, ZEN
Intermediate	Sigmaplot, NIS Elements, Neuroexplorer, Inkscape
Beginner	ImageJ, R Studio, ChemDraw, MaxQuant, Perseus

Publications

Eckert M, Miglioni P, **Izzi F** et al. Proximity proteomics reveals a co-evolved LRRK2-regulatory network linked to centrosomes. *EMBO Rep* **27**, 3488–3512 (2026)

Jager SE, Pallesen LT, Lin L, **Izzi F** et al. Comparative transcriptional analysis of satellite glial cell injury response [version 1; peer review: 2 approved]. *Wellcome Open Res* 2022, **7**:156

Languages

Italian	C2, native
English	C1, fluent
German	C1, fluent

Interests and Engagement

Volleyball	Several years in semi-professional leagues
Lego	Designing, building, and combining models
Tafel	Volunteering with the Tübingen Tafel, supporting food distribution for people in need.